

ANALYZING BUSINESS PROBLEMS

a) Identify Brands that needs Promotional or Pricing Adjustments which exhibit lower sales performance but higher profit margins.

In [180...

supplier_performance.columns

Out[180...

Index(['vendor_id', 'vendor_name', 'brand_id', 'brand_name', 'volume',
 'actual_price_per_unit', 'purchased_price_per_unit',
 'total_qty_purchased', 'total_purchases_amount',
 'selling_price_per_unit', 'total_qty_sold', 'total_sales_amount',
 'total_sales_price', 'total_excise_tax', 'total_freight_costs',
 'gross_profit_loss_amount', 'gross_profit_loss_status',
 'gross_profit_margin', 'net_profit_loss_amount',
 'net_profit_loss_status', 'net_profit_margin', 'stock_turnover',
 'sales_to_purchase_price_ratio'],
 dtype='object')

In [247...] brand_performance = supplier_performance.groupby('brand_name').agg({'total_sales_amount': 'sum', 'gross_profit_margin': 'mean'}).reset_index()

brand_performance

Out[247...

	brand_name	total_sales_amount	gross_profit_margin
0	(RI) 1	21519.09	18.06
1	.nparalleled Svgn Blanc	1094.63	29.98
2	10 Span Cab Svgn CC	2703.89	20.94
3	10 Span Chard CC	3325.56	27.81
4	10 Span Pnt Gris Monterey Cy	2082.22	32.23
...	...	...	...
7702	Zorvino Vyds Sangiovese	10579.03	29.53
7703	Zuccardi Q Malbec	1639.18	23.98
7704	Zum Rsl	10857.34	32.68
7705	Zwack Liqueur	227.88	16.65
7706	von Buhl Jazz Rsl	1359.11	90.77

7707 rows × 3 columns

Now, we need to target those brands who have a high profit margin but lower sales. For this we will create a threshold. The threshold are:

- Identify **total sales amount** below **15%** threshold
- Identify **profit margin** above **85%** threshold

In [220...

low_sales_threshold = brand_performance['total_sales_amount'].quantile(0.15)

```
In [ ]: high_margin_threshold = brand_performance['gross_profit_margin'].quantile(0.85)
```

```
In [222... print(low_sales_threshold)
print(high_margin_threshold)
```

577.4759999999998
65.04399999999998

Hence, we will target those brands who have total sales amount less than 577.74 but have gross profit margin higher than 65.04 %

```
In [230... target_brands = brand_performance[(brand_performance['total_sales_amount'] <= low_sales_threshold) & (brand_performance['gross_profit_margin'] >= high_margin_threshold)].sort_values('total_sa
```

```
In [235... target_brands
```

Out[235...

	brand_name	total_sales_amount	gross_profit_margin
6199	Santa Rita Organic Svgn Bl	9.99	66.47
2369	Debauchery Pnt Nr	11.58	65.98
2070	Concannon Glen Ellen Wh Zin	15.95	83.45
2188	Crown Royal Apple	27.86	89.81
6237	Sauza Sprklg Wild Berry Marg	27.96	82.15
...	...	...	...
57	A Bichot Clos Marechaudes	539.94	67.74
6245	Sbragia Home Ranch Merlot	549.75	66.44
3326	Goulee Cos d'Estournel 10	558.87	69.43
1204	Canaletto Pnt Grigio Venezia	561.37	87.91
4819	Merriam Cuvee Pnt Nr RRV	574.77	66.34

193 rows × 3 columns

```
In [ ]: # For better visualization
brand_performance = brand_performance[brand_performance['total_sales_amount'] < 10000]
```

```
In [290... plt.figure(figsize=(12,6))

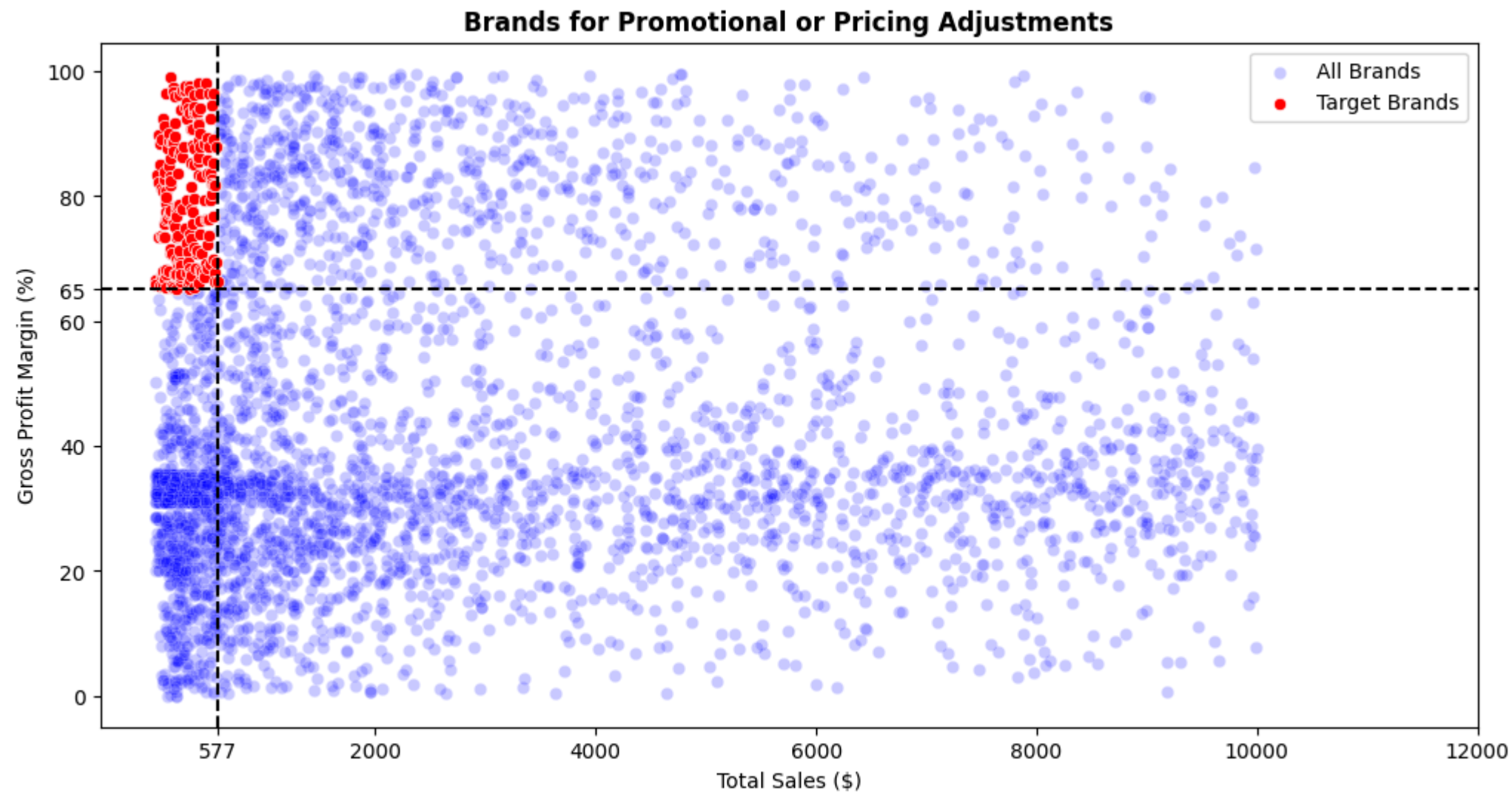
ax = sns.scatterplot(data=brand_performance, x='total_sales_amount', y='gross_profit_margin', color='blue', label = 'All Brands', alpha = 0.2)
bx = sns.scatterplot(data=target_brands, x='total_sales_amount', y='gross_profit_margin', color='red', label='Target Brands')

plt.axhline(high_margin_threshold, linestyle = '--', color='black', label = "High Margin Threshold")
plt.axvline(low_sales_threshold, linestyle = '--', color='black', label = "Low Sales Threshold")

ax.set_xticks([tick for tick in ax.get_xticks() if tick > 0] + [low_sales_threshold])
ax.set_yticks([tick for tick in ax.get_yticks() if (tick >= 0) & (tick <= 100)] + [high_margin_threshold])

plt.xlabel("Total Sales ($)")
plt.ylabel("Gross Profit Margin (%)")
plt.title("Brands for Promotional or Pricing Adjustments", fontweight = 'bold')

plt.show()
```



- Above in the red are the target brands who have low sales but high profit margin.
-

b) Which suppliers and brands demonstrate the highest sales performance?

```
In [301... top_suppliers = supplier_performance.groupby('vendor_name')['total_sales_amount'].sum().nlargest(10).reset_index()
top_brands = supplier_performance.groupby('brand_name')['total_sales_amount'].sum().nlargest(10).reset_index()
```

```
In [302... top_suppliers
```

Out[302...

	vendor_name	total_sales_amount
0	DIAGEO NORTH AMERICA INC	67990099.42
1	MARTIGNETTI COMPANIES	39330359.36
2	PERNOD RICARD USA	32063196.19
3	JIM BEAM BRANDS COMPANY	31423020.46
4	BACARDI USA INC	24854817.14
5	CONSTELLATION BRANDS INC	24218745.65
6	E & J GALLO WINERY	18399899.46
7	BROWN-FORMAN CORP	18247230.65
8	ULTRA BEVERAGE COMPANY LLP	16502544.31
9	M S WALKER INC	14706458.51

In [303...

top_brands

Out[303...

	brand_name	total_sales_amount
0	Jack Daniels No 7 Black	7964746.76
1	Tito's Handmade Vodka	7399657.58
2	Grey Goose Vodka	7209608.06
3	Capt Morgan Spiced Rum	6356320.62
4	Absolut 80 Proof	6244752.03
5	Jameson Irish Whiskey	5715759.69
6	Ketel One Vodka	5070083.56
7	Baileys Irish Cream	4150122.07
8	Kahlua	3604858.66
9	Tanqueray	3456697.90

In [355...

```
def format_number(value):
    if value >= 1000000:
        return f"{value / 1000000 : .2f}M"
    elif value >= 1000:
        return f"{value / 1000 : .2f}K"
    else:
        return str(value)
```

In [359...

```
plt.figure(figsize=(14,5))

plt.subplot(1,2,1)

ax = sns.barplot(data=top_brands, x='total_sales_amount', y='brand_name', palette="Reds_r")

plt.xlabel("Total Sales Amount", fontsize=12)
plt.ylabel("Brand Name", fontsize=12)
```

```
plt.title("Top 10 Brands by Sales", fontsize=12, fontweight='bold')

for bars in ax.containers:
    ax.bar_label(bars, labels=[format_number(bar.get_width()) for bar in bars], label_type='center', color = 'black', fontsize = 10, fontweight='bold')

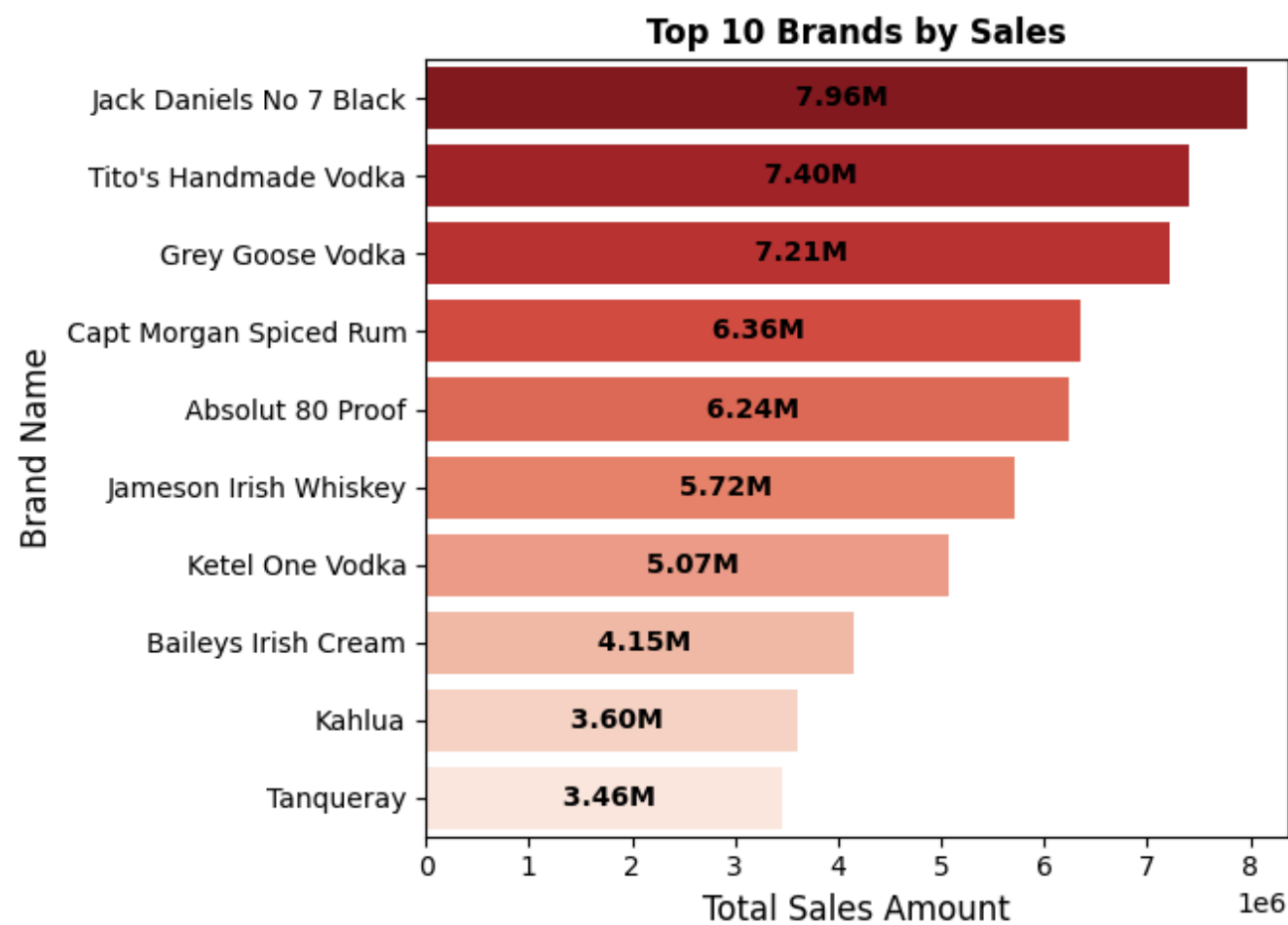
plt.subplot(1,2,2)

bx = sns.barplot(data=top_suppliers, x='total_sales_amount', y='vendor_name', palette='Greens_r')

plt.xlabel("Total Sales Amount", fontsize=12)
plt.ylabel("Supplier Name", fontsize=12)
plt.title("Top 10 Suppliers by Sales", fontsize=12, fontweight='bold')

for bars in bx.containers:
    bx.bar_label(bars, labels = [format_number(bar.get_width()) for bar in bars], label_type='center', color = 'black', fontsize = 10, fontweight = 'bold')

plt.tight_layout()
plt.show()
```



- The Red Bar Plot shows Top 10 Brands by Sales whereas Green Bar Plot shows Top 10 Suppliers by Sales

c) Which suppliers contribute the most to total purchase dollars?

In [360... supplier_performance.columns

```
Out[360...] Index(['vendor_id', 'vendor_name', 'brand_id', 'brand_name', 'volume',  
      'actual_price_per_unit', 'purchased_price_per_unit',  
      'total_qty_purchased', 'total_purchases_amount',  
      'selling_price_per_unit', 'total_qty_sold', 'total_sales_amount',  
      'total_sales_price', 'total_excise_tax', 'total_freight_costs',  
      'gross_profit_loss_amount', 'gross_profit_loss_status',  
      'gross_profit_margin', 'net_profit_loss_amount',  
      'net_profit_loss_status', 'net_profit_margin', 'stock_turnover',  
      'sales_to_purchase_price_ratio'],  
      dtype='object')
```

```
In [361...] supplier_performance.groupby('vendor_name')['total_purchases_amount'].sum().nlargest(10).reset_index()
```

Out[361...]

	vendor_name	total_purchases_amount
0	DIAGEO NORTH AMERICA INC	50097226.10
1	MARTIGNETTI COMPANIES	25502095.85
2	PERNOD RICARD USA	23851164.13
3	JIM BEAM BRANDS COMPANY	23494304.34
4	BACARDI USA INC	17432020.29
5	CONSTELLATION BRANDS INC	15273708.08
6	BROWN-FORMAN CORP	13238939.15
7	E & J GALLO WINERY	12068539.22
8	ULTRA BEVERAGE COMPANY LLP	11167081.62
9	M S WALKER INC	9764312.60

```
In [393...] suppliers = supplier_performance.groupby('vendor_name').agg({'total_purchases_amount': 'sum', 'total_sales_amount': 'sum', 'gross_profit_loss_amount': 'sum'}).reset_index()  
  
suppliers.head()
```

Out[393...]

	vendor_name	total_purchases_amount	total_sales_amount	gross_profit_loss_amount
0	ADAMBIA IMPORTS INTL INC	446.16	704.53	258.37
1	ALISA CARR BEVERAGES	25698.12	104470.94	78772.82
2	ALTAMAR BRANDS LLC	11706.20	15706.81	4000.61
3	AMERICAN SPIRITS EXCHANGE	934.08	1511.16	577.08
4	AMERICAN VINTAGE BEVERAGE	104435.68	139603.53	35167.85

```
In [394...] suppliers['purchase_percentage_contribution'] = (suppliers['total_purchases_amount'] / suppliers['total_purchases_amount'].sum() * 100).round(2)  
  
suppliers.sort_values('purchase_percentage_contribution', ascending=False)
```


Out[394...

	vendor_name	total_purchases_amount	total_sales_amount	gross_profit_loss_amount	purchase_percentage_contribution
25	DIAGEO NORTH AMERICA INC	50097226.10	67990099.42	17892873.32	16.06
57	MARTIGNETTI COMPANIES	25502095.85	39330359.36	13828263.51	8.17
68	PERNOD RICARD USA	23851164.13	32063196.19	8212032.06	7.65
46	JIM BEAM BRANDS COMPANY	23494304.34	31423020.46	7928716.12	7.53
6	BACARDI USA INC	17432020.29	24854817.14	7422796.85	5.59
...	...	...	...	...	...
101	THE IMPORTED GRAPE LLC	12379.45	18703.65	6324.20	0.00
107	UNCORKED	118.74	176.94	58.20	0.00
108	VINEDREA WINES LLC	4657.60	11385.60	6728.00	0.00
105	TY KU LLC	13068.46	21476.58	8408.12	0.00
113	WALPOLE MTN VIEW WINERY	9292.31	11483.88	2191.57	0.00

119 rows × 5 columns

In [395...

```
suppliers['total_purchases_amount'] = suppliers['total_purchases_amount'].apply(format_number)
suppliers['total_sales_amount'] = suppliers['total_sales_amount'].apply(format_number)
suppliers['gross_profit_loss_amount'] = suppliers['gross_profit_loss_amount'].apply(format_number)
```

In [420...

```
top_suppliers = suppliers.sort_values('purchase_percentage_contribution', ascending=False).head(10)

top_suppliers['cumulative_percentage'] = top_suppliers['purchase_percentage_contribution'].cumsum()

top_suppliers
```

Out[420...

	vendor_name	total_purchases_amount	total_sales_amount	gross_profit_loss_amount	purchase_percentage_contribution	cumulative_percentage
25	DIAGEO NORTH AMERICA INC	50.10M	67.99M	17.89M	16.06	16.06
57	MARTIGNETTI COMPANIES	25.50M	39.33M	13.83M	8.17	24.23
68	PERNOD RICARD USA	23.85M	32.06M	8.21M	7.65	31.88
46	JIM BEAM BRANDS COMPANY	23.49M	31.42M	7.93M	7.53	39.41
6	BACARDI USA INC	17.43M	24.85M	7.42M	5.59	45.00
20	CONSTELLATION BRANDS INC	15.27M	24.22M	8.95M	4.90	49.90
11	BROWN-FORMAN CORP	13.24M	18.25M	5.01M	4.24	54.14
30	E & J GALLO WINERY	12.07M	18.40M	6.33M	3.87	58.01
106	ULTRA BEVERAGE COMPANY LLP	11.17M	16.50M	5.34M	3.58	61.59
53	M S WALKER INC	9.76M	14.71M	4.94M	3.13	64.72

In [619...

```
plt.figure(figsize=(10,8))

# Bar chart
ax = sns.barplot(data=top_suppliers, x = 'vendor_name', y='purchase_percentage_contribution', palette='mako')
```

```
ax.set_xticklabels(ax.get_xticklabels(), rotation=90)

for bars in ax.containers:
    ax.bar_label(bars, labels=[f"{bar.get_height(): .2f}%" for bar in bars], label_type='edge', color = "black")

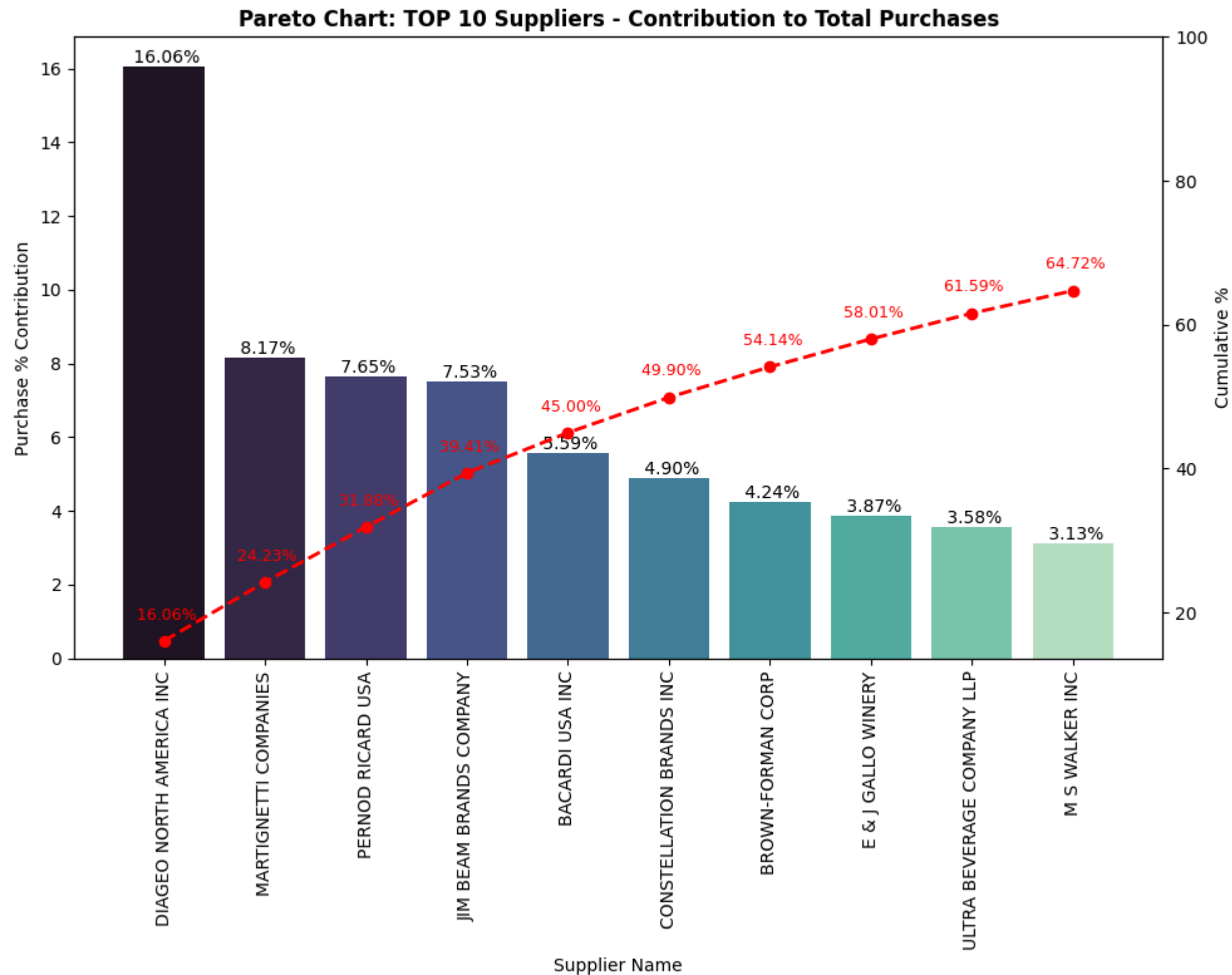
# Cumulative graph
ax2 = ax.twinx()
ax2.plot(top_suppliers['vendor_name'], top_suppliers['cumulative_percentage'], color = 'red', marker = 'o', linestyle = 'dashed', linewidth = 2, label = 'Cumulative %')

for i, val in enumerate(top_suppliers['cumulative_percentage']):
    ax2.text(i, val+3, f"{val: .2f}%", ha='center', color='red', fontsize=9)

ax.set_ylabel("Purchase % Contribution")
ax2.set_ylabel("Cumulative %")
ax.set_xlabel("Supplier Name")

ax2.set_yticks([20, 40, 60, 80, 100])

plt.title("Pareto Chart: TOP 10 Suppliers - Contribution to Total Purchases", fontweight = 'bold')
plt.tight_layout()
plt.show()
```

- We can observe that top 10 suppliers contributed to around 65% of total purchases made.

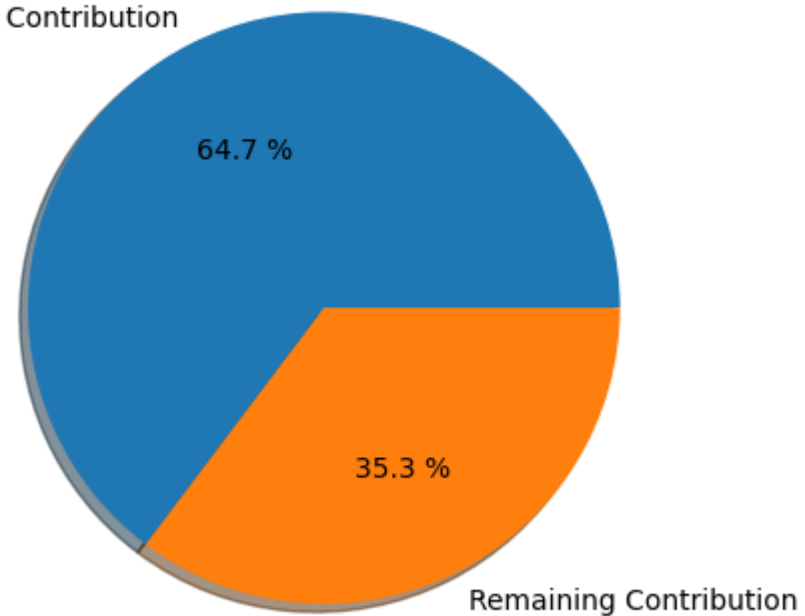
d) Compare the total procurement value of the top 10 suppliers with that of all remaining suppliers.

```
In [483... top_10_contribution = top_suppliers['purchase_percentage_contribution'].sum()
remaining_contribution = 100-top_10_contribution
contribution = [top_10_contribution, remaining_contribution]
contribution = [float(x) for x in contribution]
categories = ['Top 10 Contribution', 'Remaining Contribution']
```

```
In [484... plt.pie(contribution, labels = categories, autopct='%1.1f %%', shadow = True)
plt.title("Pie Chart: Contribution", fontsize=12, fontweight='bold')
plt.show()
```

Pie Chart: Contribution

Top 10 Contribution



```
In [572... top_10_suppliers = list(top_suppliers['vendor_name'].values)
purchase_contribution = [float(x) for x in top_suppliers['purchase_percentage_contribution'].values]
total_contribution = sum(purchase_contribution)
remaining_contribution = 100 - total_contribution

# Append Other Suppliers Category. This not affects overall sum
top_10_suppliers.append("Other Suppliers")
purchase_contribution.append(remaining_contribution)

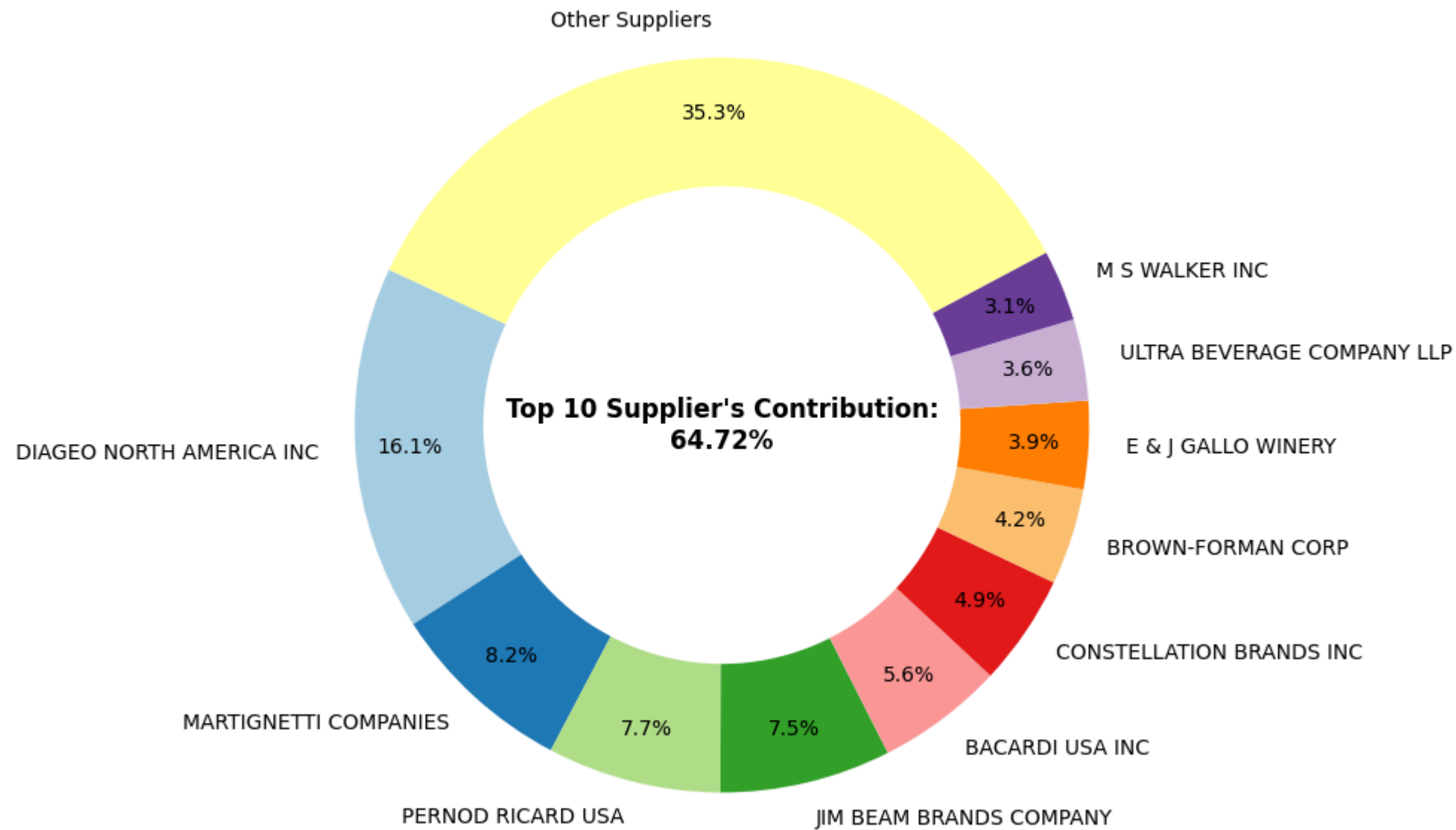
# Donut Chart
fig, ax = plt.subplots(figsize = (8,8))
wedges, texts, autotexts = ax.pie(purchase_contribution, labels=top_10_suppliers, autopct='%1.1f%%', startangle=155, pctdistance=0.85, colors=plt.cm.Paired.colors)

# Centre Circle
centre_circle = plt.Circle((0, 0), 0.65, fc='white')
ax.add_artist(centre_circle)

# Add annotation inside the donut
plt.text(0, 0, f"Top 10 Supplier's Contribution:\n{total_contribution:.2f}%", fontsize=12, fontweight='bold', ha='center', va='center')

plt.title("SUPPLIERS PROCUREMENT CONTRIBUTION", fontweight = 'bold')
plt.show()
```

SUPPLIERS PROCUREMENT CONTRIBUTION



- 64.7% of Procurement are made by Top 10 suppliers whereas Other suppliers contribute to only 35.28%.

e) Does purchasing in bulk reduce the unit price, and what is the optimal purchase volume for cost savings?

In [574...

```
supplier_performance['avg_purchase_price'] = supplier_performance['total_purchases_amount'] / supplier_performance['total_qty_purchased']
supplier_performance.head(3)
```

Out[574...

	vendor_id	vendor_name	brand_id	brand_name	volume	actual_price_per_unit	purchased_price_per_unit	total_qty_purchased	total_purchases_amount	selling_price_per_unit	total_qty_sold	total_sales_amount
0	1128	BROWN-FORMAN CORP	1233	Jack Daniels No 7 Black	1750.0	36.99	26.27	145080	3811251.60	34.99	142049.0	5101919.51
1	4425	MARTIGNETTI COMPANIES	3405	Tito's Handmade Vodka	1750.0	28.99	23.19	164038	3804041.22	27.95	160247.0	4819073.49
2	17035	PERNOD RICARD USA	8068	Absolut 80 Proof	1750.0	24.99	18.24	187407	3418303.68	23.49	187140.0	4538120.60

In [579...

```
supplier_performance['order_size'] = pd.qcut(supplier_performance['total_qty_purchased'], q=3, labels=['Small', 'Medium', 'Large'])
supplier_performance.head(3)
```

Out[579...

	vendor_id	vendor_name	brand_id	brand_name	volume	actual_price_per_unit	purchased_price_per_unit	total_qty_purchased	total_purchases_amount	selling_price_per_unit	total_qty_sold	total_sales_amount
0	1128	BROWN-FORMAN CORP	1233	Jack Daniels No 7 Black	1750.0	36.99	26.27	145080	3811251.60	34.99	142049.0	5101919.51
1	4425	MARTIGNETTI COMPANIES	3405	Tito's Handmade Vodka	1750.0	28.99	23.19	164038	3804041.22	27.95	160247.0	4819073.49
2	17035	PERNOD RICARD USA	8068	Absolut 80 Proof	1750.0	24.99	18.24	187407	3418303.68	23.49	187140.0	4538120.60

In [582...

```
supplier_performance['order_size'].value_counts().reset_index()
```

Out[582...

	order_size	count
0	Small	3010
1	Large	2963
2	Medium	2917

In [585...

```
supplier_performance.groupby('order_size')['avg_purchase_price'].mean().reset_index()
```

Out[585...

	order_size	avg_purchase_price
0	Small	41.395515
1	Medium	15.421172
2	Large	10.744448

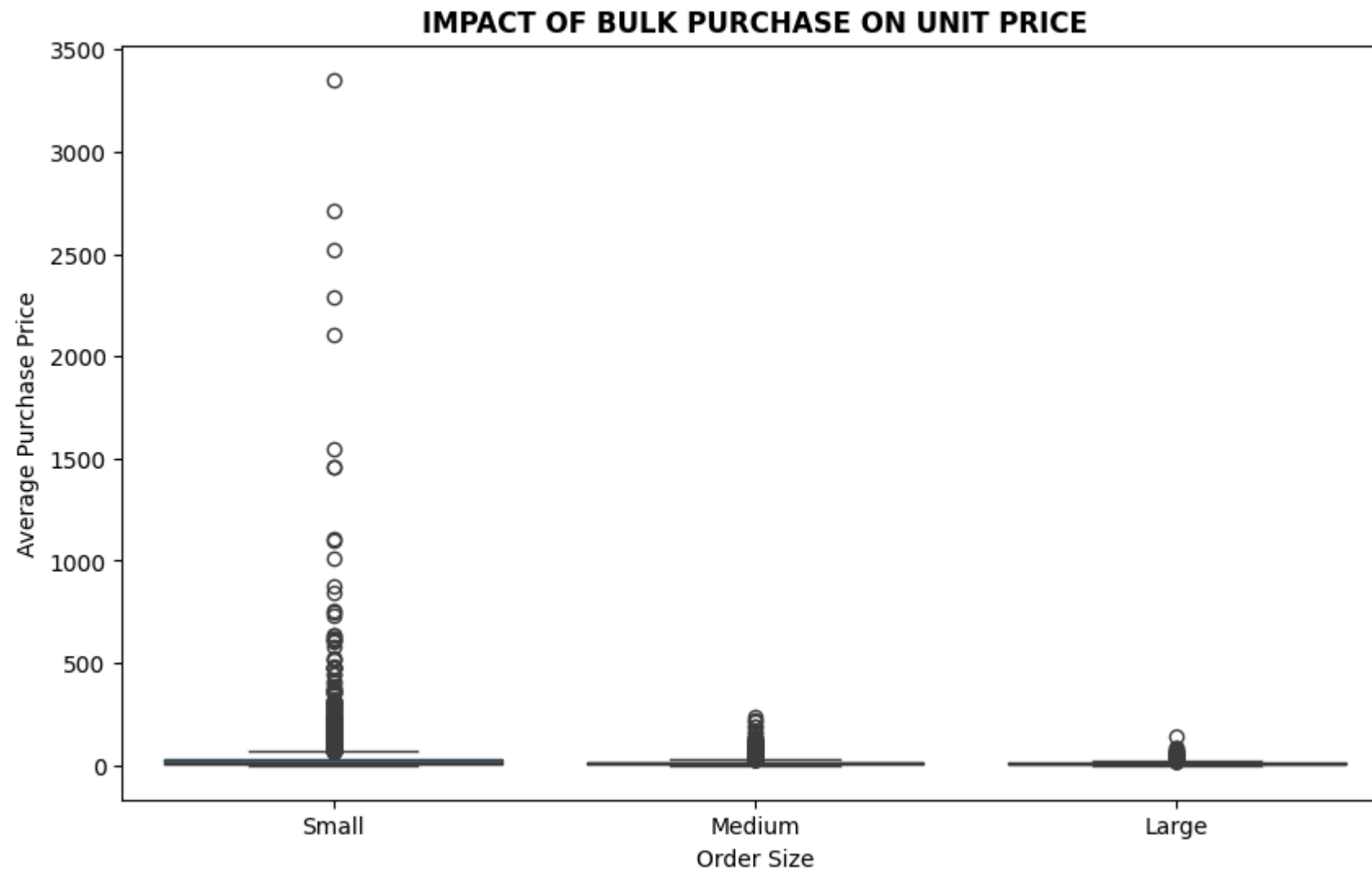
In [598...

```
plt.figure(figsize=(10,6))

sns.boxplot(data = supplier_performance, x='order_size', y='avg_purchase_price')

plt.title("IMPACT OF BULK PURCHASE ON UNIT PRICE", fontweight = 'bold')
plt.xlabel("Order Size")
```

```
plt.ylabel("Average Purchase Price")
plt.show()
```



- The box plot reveals that the Small category has numerous outliers, whereas the Large category exhibits very few. This suggests that the Large category has a narrower range of values, supporting the idea that bulk purchasing helps reduce the average purchase price per unit.
- The price difference between Small and Large orders is significant, with a 74% reduction in unit cost — showing that bulk purchasing greatly lowers the price per unit.
- The LARGE category represents the optimal purchase volume for bulk buying.

f) Which suppliers have low inventory turnover, indicating excess stock and slow-moving products?

```
In [612... low_inv_turnover = supplier_performance[supplier_performance['stock_turnover'] < 1].groupby('vendor_name')['stock_turnover'].mean().reset_index().sort_values('stock_turnover').head(10)
low_inv_turnover
```

Out[612...

	vendor_name	stock_turnover
0	ALISA CARR BEVERAGES	0.620000
36	HIGHLAND WINE MERCHANTS LLC	0.710000
60	PARK STREET IMPORTS LLC	0.750000
19	Circa Wines	0.755385
41	KLIN SPIRITS LLC	0.760000
26	Dunn Wine Brokers	0.767500
15	CENTEUR IMPORTS LLC	0.772500
78	SMOKY QUARTZ DISTILLERY LLC	0.780000
90	TAMWORTH DISTILLING	0.800000
91	THE IMPORTED GRAPE LLC	0.808182

In [705...

```
plt.figure(figsize=(7,6))

ax = sns.barplot(data=low_inv_turnover, x='stock_turnover', y='vendor_name', palette='Greens_r')

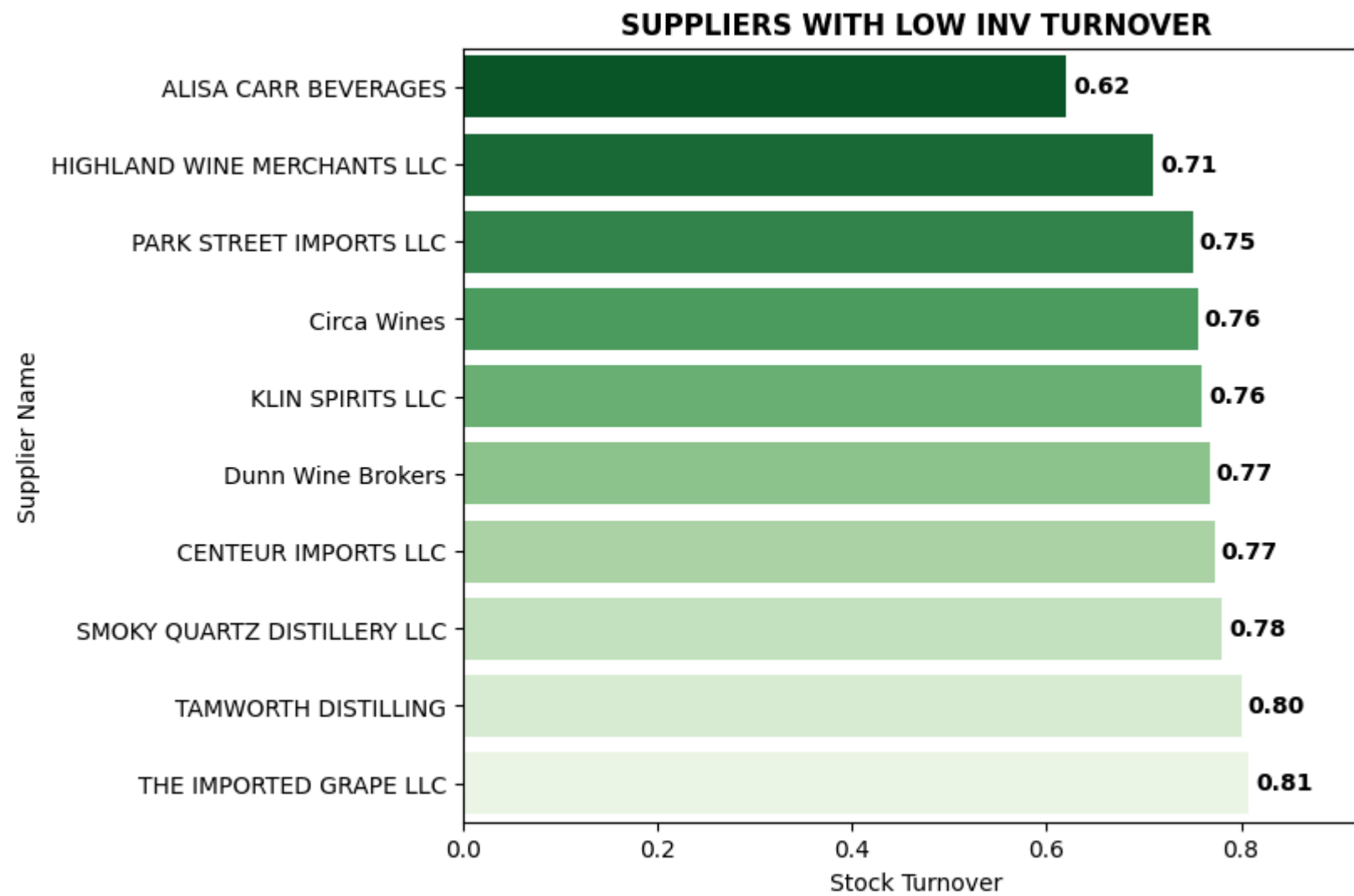
for bars in ax.containers:
    ax.bar_label(bars, fmt='%.2f', label_type='edge', color = 'black', fontsize = 10, fontweight='bold', padding=3)

# Gives min and max x-axis values
xmin, xmax = ax.get_xlim()

# Extend xmax by 10%:
ax.set_xlim(xmin, xmax * 1.1)

plt.xlabel('Stock Turnover')
plt.ylabel('Supplier Name')
plt.title('SUPPLIERS WITH LOW INV TURNOVER', fontweight = 'bold')

plt.show()
```

- Above are Top 10 Suppliers with the lowest inventory turnover.

g) Which suppliers are responsible for the largest share of unsold inventory value?

```
In [661...] supplier_performance['unsold_inv_value'] = (supplier_performance['total_qty_purchased'] - supplier_performance['total_qty_sold']) * supplier_performance['purchased_price_per_unit']
format_number(supplier_performance['unsold_inv_value'].sum())
```

Out[661...] ' 2.67M'

VinSpire currently has \$2.6 million worth of capital tied up in inventory. Let's observe top contributors to this.

```
In [678...] unsold_inv = supplier_performance.groupby('vendor_name')['unsold_inv_value'].sum().reset_index().nlargest(10,'unsold_inv_value')
unsold_inv
```

Out[678...

	vendor_name	unsold_inv_value
25	DIAGEO NORTH AMERICA INC	722209.05
46	JIM BEAM BRANDS COMPANY	554665.63
68	PERNOD RICARD USA	470625.61
116	WILLIAM GRANT & SONS INC	401960.83
30	E & J GALLO WINERY	228282.61
79	SAZERAC CO INC	198436.41
11	BROWN-FORMAN CORP	177733.74
20	CONSTELLATION BRANDS INC	133617.62
61	MOET HENNESSY USA INC	126477.70
77	REMY COINTREAU USA INC	118598.15

In [707...

```
plt.figure(figsize=(8,6))

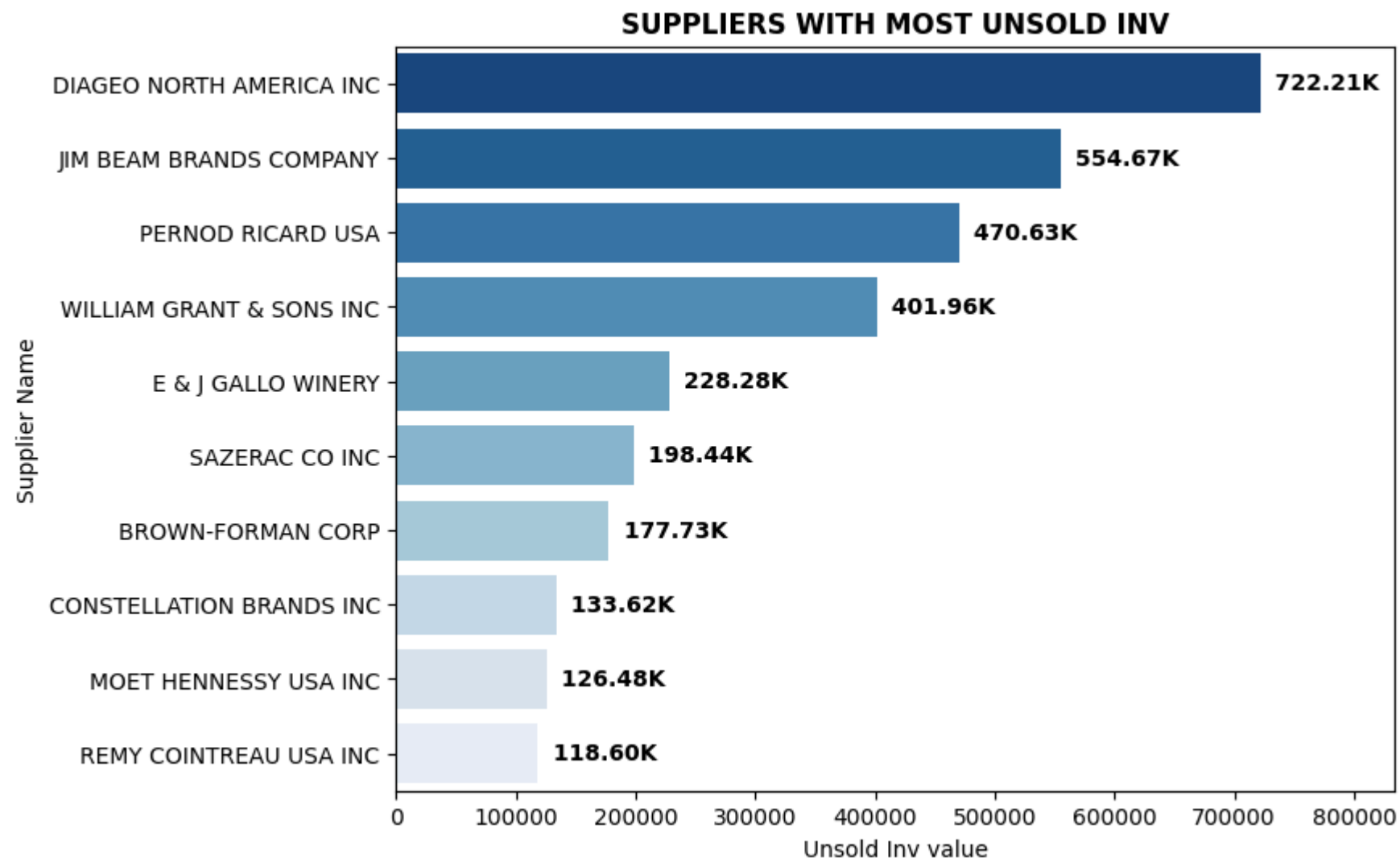
ax = sns.barplot(data=unsold_inv, x='unsold_inv_value', y='vendor_name', palette='Blues_r')

for bars in ax.containers:
    ax.bar_label(bars, labels=[format_number(bar.get_width()) for bar in bars], label_type='edge', color = 'black', fontsize = 10, fontweight='bold', padding=3)

xmin, xmax = ax.get_xlim()
ax.set_xlim(xmin, xmax * 1.1)

plt.xlabel('Unsold Inv value')
plt.ylabel('Supplier Name')
plt.title('SUPPLIERS WITH MOST UNSOLD INV', fontweight = 'bold')

plt.show()
```

- These are Top 10 Suppliers currently contributing to \$2.6 million worth of capital tied up in inventory.

h) Identify the 95% confidence intervals for profit margins of top-performing and low-performing suppliers.

```
In [ ]: # Set a threshold to identify top and bottom suppliers by sales amount.
```

```
high_threshold = supplier_performance['total_sales_amount'].quantile(0.75)
low_threshold = supplier_performance['total_sales_amount'].quantile(0.25)
```

```
In [737... print(high_threshold)
print(low_threshold)
```

```
39520.44
1359.1625000000001
```

Suppliers having sales higher than high_threshold are top_performing whereas suppliers having sales less than low_threshold are low performing.

```
In [733... top_suppliers = supplier_performance[supplier_performance['total_sales_amount'] >= high_threshold]['gross_profit_margin']
bottom_suppliers = supplier_performance[supplier_performance['total_sales_amount'] <= low_threshold]['gross_profit_margin']
```

```
In [734... top_suppliers
```

```
Out[734...] 0      25.30
            1      21.06
            2      24.68
            3      27.14
            4      28.41
            ...
            4078    79.68
            4287    85.78
            5694    93.09
            5946    95.01
            6126    94.27
Name: gross_profit_margin, Length: 2224, dtype: float64
```

```
In [735...] bottom_suppliers
```

```
Out[735...] 6942      4.11
            6977      6.15
            7047     12.01
            7050      1.68
            7083      7.24
            ...
            11096    83.45
            11097    96.44
            11098    25.25
            11099    98.97
            11100    99.17
Name: gross_profit_margin, Length: 2223, dtype: float64
```

To find Confidence Interval, the formula is:

CI = mean ± score * (std/(√n))

```
In [789...] from scipy.stats import norm, t
```

Percent Point Function (.ppf) - When you have confidence and need to find z/t score

Cumulative Distribution Function (.cdf) - When you have z/t score and need to confidence

```
In [ ]: def confidence_interval(data, confidence = 0.95):
        mean = np.mean(data)
        std = np.std(data, ddof=1)      # Degree of freedom = 1 (For sample)
        n = len(data)

        if n < 30:
            score = t.ppf((1+confidence) / 2, df = n-1)
        else:
            score = norm.ppf((1+confidence) / 2)

        margin_of_error = score * (std / np.sqrt(n))
        lower_bound = mean - margin_of_error
        upper_bound = mean + margin_of_error

        return round(mean, 2), round(lower_bound, 2), round(upper_bound, 2)
```

```
In [791...] confidence_interval(top_suppliers)
```

```
Out[791...] (np.float64(31.31), np.float64(30.89), np.float64(31.74))
```

```
In [830... confidence_interval(bottom_suppliers)
```

```
Out[830... (np.float64(41.74), np.float64(40.69), np.float64(42.78))
```

- **Top_suppliers** have an average profit of 31.31% and we are 95% sure that their Gross Profit Margin lies between 30.89% and 31.74%
- **Bottom_suppliers** have an average profit of 41.74% and we are 95% sure that their Gross Profit Margin lies between 40.69% and 42.78%

```
In [807... top_mean, top_lower, top_upper = confidence_interval(top_suppliers)
bottom_mean, bottom_lower, bottom_upper = confidence_interval(bottom_suppliers)

fig, axes = plt.subplots(1, 2, figsize=(14, 6), sharey=True)

print(f"Top_Suppliers 95% CI: ({top_lower:.2f}, {top_upper:.2f}), Mean: {top_mean}")
print(f"Bottom_Suppliers 95% CI: ({bottom_lower:.2f}, {bottom_upper:.2f}), Mean: {bottom_mean}")

# Top Suppliers

sns.histplot(top_suppliers, kde=True, color='royalblue', edgecolor='black', bins=30, ax=axes[0])
axes[0].axvline(top_lower, color='red', linestyle='--', label=f'Lower 95% CI: {top_lower}')
axes[0].axvline(top_upper, color='red', linestyle='--', label=f'Upper 95% CI: {top_upper}')
axes[0].axvline(top_mean, color='red', linestyle='-', label=f'Mean: {top_mean}')
axes[0].set_title('Top Suppliers')
axes[0].set_xlabel('Gross Profit Margin (%)')
axes[0].set_ylabel('Frequency')
axes[0].legend()

# Bottom Suppliers

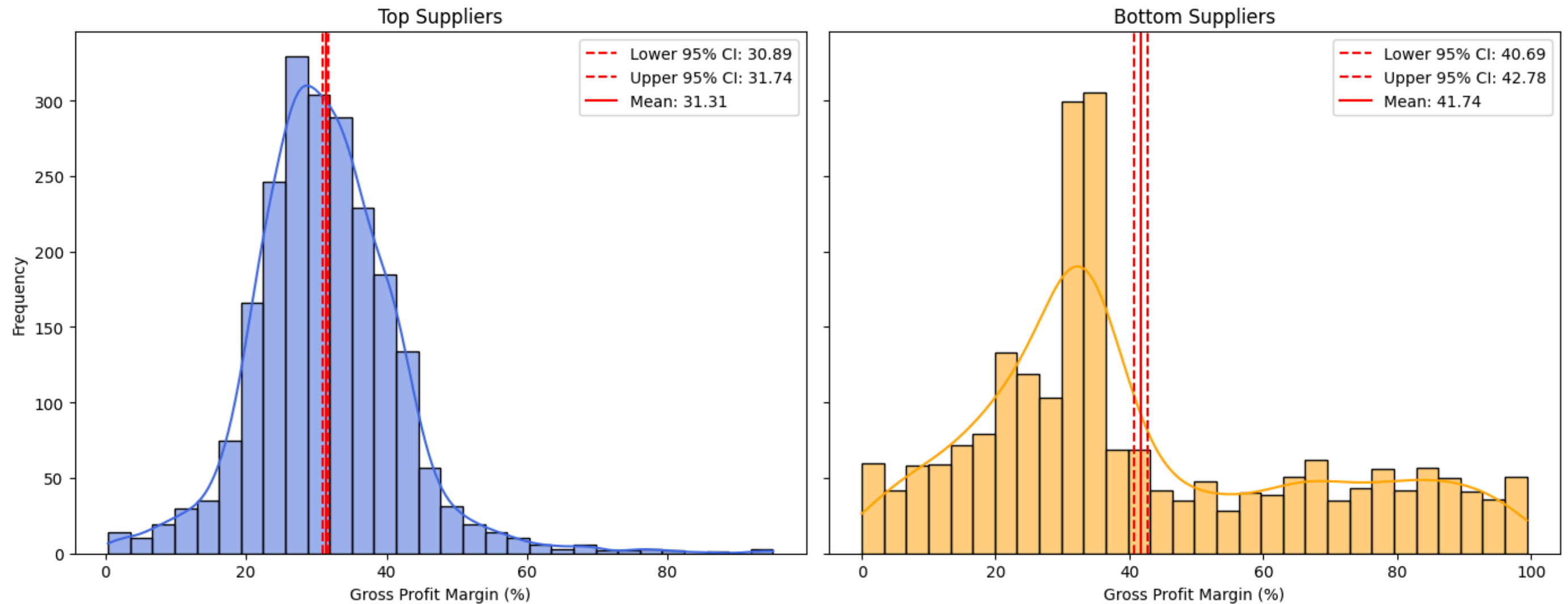
sns.histplot(bottom_suppliers, kde=True, color='orange', edgecolor='black', bins=30, ax=axes[1])
axes[1].axvline(bottom_lower, color='red', linestyle='--', label=f'Lower 95% CI: {bottom_lower}')
axes[1].axvline(bottom_upper, color='red', linestyle='--', label=f'Upper 95% CI: {bottom_upper}')
axes[1].axvline(bottom_mean, color='red', linestyle='-', label=f'Mean: {bottom_mean}')
axes[1].set_title('Bottom Suppliers')
axes[1].set_xlabel('Gross Profit Margin (%)')
axes[1].legend()

plt.suptitle('Gross Profit Margin Distribution', fontsize = 20, fontweight='bold')
plt.tight_layout()
plt.show()
```

Top_Suppliers 95% CI: (30.89, 31.74), Mean: 31.31

Bottom_Suppliers 95% CI: (40.69, 42.78), Mean: 41.74

Gross Profit Margin Distribution



OBSERVATION:

- Low-performing suppliers have a higher gross profit margin, with a 95% confidence interval ranging from 40.69% to 42.78%, compared to 30.89% to 31.74% for top-performing suppliers.
- This indicates that low-volume suppliers may be leveraging premium pricing, operating in niche markets, or benefiting from lower operational costs.
- Top-performing suppliers, while driving higher sales, operate with tighter margins. To enhance profitability, they could consider strategic price adjustments, cost optimization, or value-added bundling to increase margin without losing volume.
- Low-performing suppliers, despite having higher margins, may struggle with limited reach or market visibility. They should focus on competitive pricing, stronger marketing efforts, or improving distribution to boost sales volume.

i) Is there a significant difference in profit margins between top-performing and low-performing suppliers?

HYPOTHESIS:

- **H0 (Null Hypothesis):** There is no significant difference in the mean profit margins of top-performing and low-performing suppliers.
- **H1 (Alternative Hypothesis):** The mean profit margins of top-performing and low-performing suppliers are significantly different.

In [808... `from scipy.stats import ttest_ind`

```
In [ ]: high_threshold = supplier_performance['total_sales_amount'].quantile(0.75)
low_threshold = supplier_performance['total_sales_amount'].quantile(0.25)

top_suppliers = supplier_performance[supplier_performance['total_sales_amount'] >= high_threshold]['gross_profit_margin']
bottom_suppliers = supplier_performance[supplier_performance['total_sales_amount'] <= low_threshold]['gross_profit_margin']

# Two Sample t-test

t_stat, p_val = ttest_ind(top_suppliers, bottom_suppliers, equal_var=False)
# t-stat = (mean of top - mean of bottom) / standard error

print(f"T-Stats: {t_stat:.4f}, P-Value: {p_val:.4f}")

if p_val < 0.5:
    print("Reject H0: There is a significant difference in mean profit margin between top and bottom performing suppliers")
else:
    print("Fail to Reject H0: No significant difference in mean profit margin between top and bottom performing suppliers")
```

T-Stats: -18.0578, P-Value: 0.0000

Reject H0: There is a significant difference in mean profit margin between top and bottom performing suppliers

```
In [829... t_stat = t_stat
p_val = p_val

# Degrees of freedom
n1 = len(top_suppliers)
n2 = len(bottom_suppliers)
s1 = np.std(top_suppliers, ddof=1)
s2 = np.std(bottom_suppliers, ddof=2)

df_num = (s1**2/n1 + s2**2/n2)**2
df_den = (s1**2/n1)**2 / (n1-1) + (s2**2/n2)**2 / (n2-1)
df = df_num / df_den

# Confidence Level
confidence = 0.95
alpha = 1 - confidence

# Critical t-value for two-tailed test
t_crit = t.ppf(1 - alpha/2, df)

# Prepare x range for the plot
x = np.linspace(-6, 6, 1000)
y = t.pdf(x, df)

plt.figure(figsize=(10, 6))

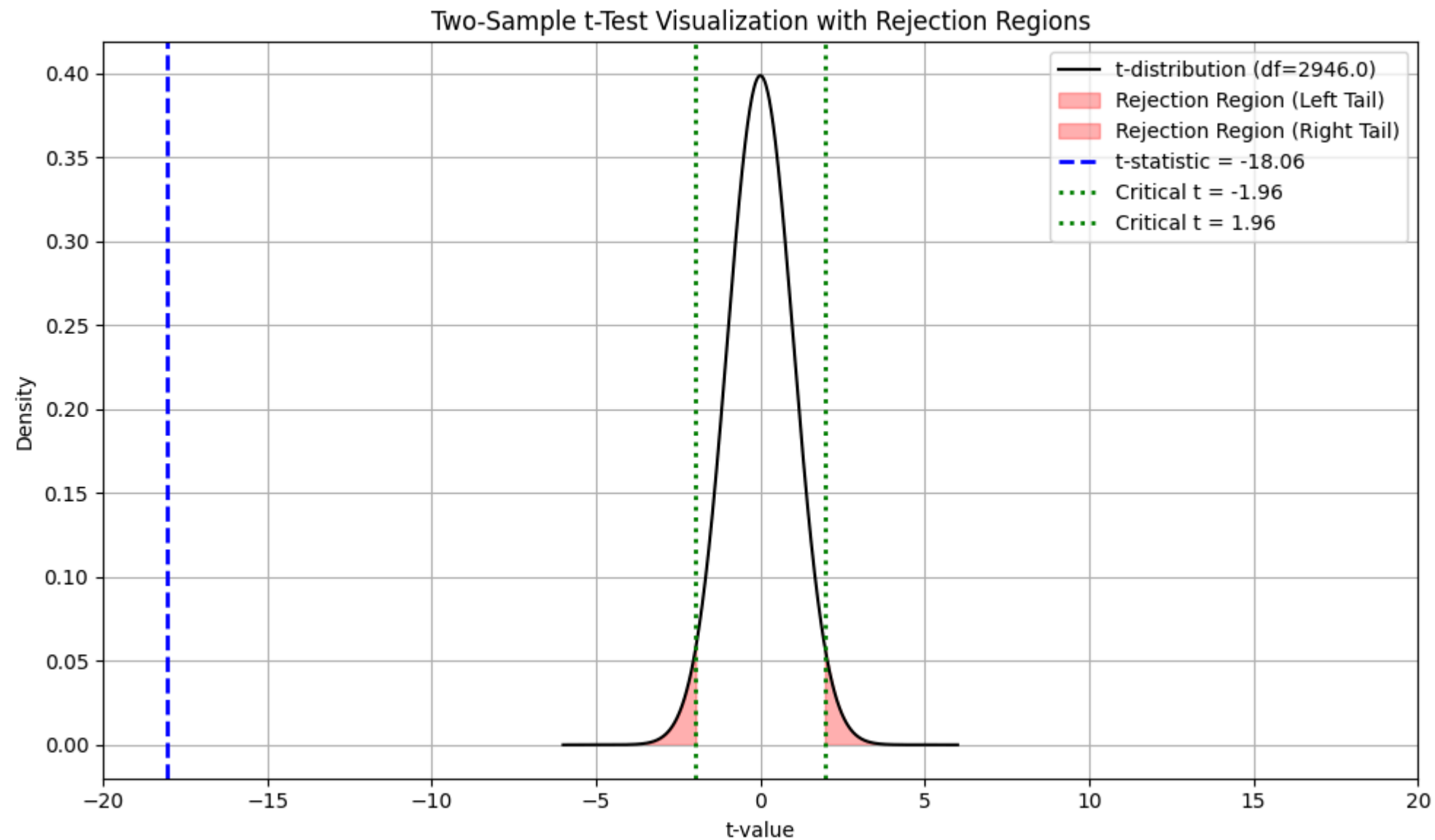
# Plot t-distribution
plt.plot(x, y, label='t-distribution (df={:.1f})'.format(df), color='black')

# Shade rejection regions
plt.fill_between(x, y, where=(x <= -t_crit), color='red', alpha=0.3, label='Rejection Region (Left Tail)')
plt.fill_between(x, y, where=(x >= t_crit), color='red', alpha=0.3, label='Rejection Region (Right Tail)')

# Mark the t-statistic from your test
plt.axvline(t_stat, color='blue', linestyle='--', linewidth=2, label=f't-statistic = {t_stat:.2f}')

# Mark critical t-values (confidence interval bounds)
plt.axvline(-t_crit, color='green', linestyle=':', linewidth=2, label=f'Critical t = {-t_crit:.2f}')
plt.axvline(t_crit, color='green', linestyle=':', linewidth=2, label=f'Critical t = {t_crit:.2f}')
```

```
plt.title('Two-Sample t-Test Visualization with Rejection Regions')
plt.xlabel('t-value')
plt.ylabel('Density')
plt.xlim(-20, 20)
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()
```



OBSERVATION:

- The t-statistic of -18.06 shows that top-performing suppliers have lower average profit margins than low-performing ones, with a difference about 18 times greater than random variation.
- The p-value (0.0000) < 0.05, meaning we reject the null hypothesis concluding there is significant difference in profit margins.
- Overall, there is a statistically significant difference in profit margins between top- and low-performing suppliers.